

CISC 327 A4 (Fall 2025)

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Repository: <https://github.com/EronChung/cisc327-library-management-a2-3151>

E2E Testing Approach

For my E2E tests, I used Playwright to automate user flow tests in a browser session.

The tests run through adding, borrowing, and removing a book, and can be found in `tests/test\_e2e.py`.

Test #	Actions	Expected Results
1	- Assert that the page title is correct (sanity check to see if on the right page) - Go to the "Add book" page - Fill in the fields and click "Add book to catalog" (this sends you back to the catalog page) - Borrow the newly added book with the patron ID `123456` - Assert that the confirmation UI element appeared	- No runtime errors - Confirmation UI element appeared for borrow
2	- Assert that the page title is correct (sanity check to see if on the right page) - Go to the "Return book" page - Fill in the fields and click "Process return" - Assert that the confirmation UI element appeared - Fill in the fields for a different book and click "Process return" - Assert that the error UI element appeared	- No runtime errors - Confirmation UI element appeared for return - Error UI element appeared for invalid return

Execute Instructions

- Clone repository
- Run `python -m venv venv` to create virtual environment
- Run `source venv/bin/activate` in Linux, or `venv\Scripts\activate` on Windows
- Run `pip install -r requirements.txt` to install dependencies
- Run `python app.py`
- In a new terminal, run `pytest tests/test\_e2e.py --headed --slowmo 500` to run E2E tests
- Run `docker build -t library-app .` to build the Docker container
- Run `docker container run -p 5000:5000 library-app` to run the container

Dockerization

Build

```
(venv) PS C:\Users\Eron\Documents\__Dev\Python\CISC327\CISC327-CMPE327-F25> docker build -t library-app .
[+] Building 16.0s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 190B
=> [internal] load metadata for docker.io/library/python:3.12
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:3.12@sha256:01f36278bcca9fb4a5bdda7edc85be44218190a9371f642d8f852f36ad9fa23d
=> => resolve docker.io/library/python:3.12@sha256:01f36278bcca9fb4a5bdda7edc85be44218190a9371f642d8f852f36ad9fa23d
=> [internal] load build context
=> => transferring context: 557.35kB
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt requirements.txt
=> CACHED [4/5] RUN pip install -r requirements.txt
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:855ec048e3e5930bb7ec36eeefa33d961c4cfa8242f2bf80bdb13f678d3de278
=> => exporting config sha256:18f94acab51eb02360e7ac799ec9e56611c8af7b17d6cca9a61c416851898a7e
=> => exporting attestation manifest sha256:61b385c7a5bebd895a98c73715d3f78490d91023fa0364b904ad51adf2e2993a
=> => exporting manifest list sha256:41145512ca59aec4e84af7a43999eeb037c712eac60e4e91239768a28222eed3
=> => naming to docker.io/library/library-app:latest
=> => unpacking to docker.io/library/library-app:latest
```

Run

```
(venv) PS C:\Users\Eron\Documents\__Dev\Python\CISC327\CISC327-CMPE327-F25> docker container run -p 5000:5000 library-app
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 103-638-886
```

Docker Hub

Push

```
(venv) PS C:\Users\Eron\Documents\__Dev\Python\CISC327\CISC327-CMPE327-F25> docker tag library-app eronchung/library-app:v1
(venv) PS C:\Users\Eron\Documents\__Dev\Python\CISC327\CISC327-CMPE327-F25> docker push eronchung/library-app:v1
The push refers to repository [docker.io/eronchung/library-app]
fecaa8094a43: Pushed
7c40a3faff76: Pushed
f88c3859cc57: Pushed
ca944efff8e2c: Pushed
53c88f1dfeb7: Pushed
858443e1534f: Pushed
bf0309231083: Pushed
f59caa67e826: Pushed
eae668646f44: Pushed
ef75e1766b83: Pushed
ff2e6e687b6c: Pushed
e74d5df456c1: Pushed
v1: digest: sha256:9662deb86571a698a1dd6eb7552efae07aafb35d86f6270c5a2020436af0075e size: 856
(venv) PS C:\Users\Eron\Documents\__Dev\Python\CISC327\CISC327-CMPE327-F25> 
```

Repositories

All repositories within the eronchung namespace.

Name	Last Pushed	
eronchung/library-app	less than a minute ago	

1-1 of 1

Pull and Run

```
(venv) PS C:\Users\Eron\Documents\__Dev\Python\CISC327\CISC327-CMPE327-F25> docker rmi eronchung/library-app:v1
Untagged: eronchung/library-app:v1
(venv) PS C:\Users\Eron\Documents\__Dev\Python\CISC327\CISC327-CMPE327-F25> docker pull eronchung/library-app:v1
v1: Pulling from eronchung/library-app
Digest: sha256:9662deb86571a698a1dd6eb7552efae07aafb35d86f6270c5a2020436af0075e
Status: Downloaded newer image for eronchung/library-app:v1
docker.io/eronchung/library-app:v1
(venv) PS C:\Users\Eron\Documents\__Dev\Python\CISC327\CISC327-CMPE327-F25> docker run -p 5000:5000 eronchung/library-app:v1
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 102-194-126
```

Challenges

Having not worked with Flask before, basically all of these assignments had me consulting various documentations a lot and watching tutorial videos on how to set up different things.